

Tube Tech

Nanotechnology is the future! We've heard this a million times before, but now it finally seems that nanotubes have a foot grounded in reality

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Tech buzzwords come and go, some more long-lasting than others. Like "convergence". But it isn't often that so many researchers in so many areas get hooked on a thing. That thing today is nanotubes. These miraculous structures are promising to deliver in so many areas that big companies—IBM, Intel, and the likes included—and small companies alike are rushing in to invest in nanotechnology in general, and nanotubes in particular.

If you missed out on what carbon nanotubes are, check out *Move Over Silicon*, *Digit*, May 2006. But here's a one-line recap: they are elegant-looking structures made of carbon (duh): hollow columns of carbon atoms. One of their most important properties is that they can be both semi-conducting and conducting; the conducting

ones conduct very well indeed, well surpassing copper and aluminium in that area. You'll find out more of their properties as you read ahead.

As our regular disclaimer goes, we can't cover everything that nanotubes are good at in this space, but we look at their applications in things ranging from displays to batteries to chip interconnects. And more.

And Nanotubes Come In...

...to save the battery. They're running out so often these days, people are getting tempted to give up on battery-powered devices altogether and take up farming. But would you believe that someone is planning on making batteries last longer by using nanotube-turbo-charged capacitors?

In case you never attended engineering college (whether you were enrolled or not), a capacitor is a crucial component of almost anything electronic, and it holds charge, which it can release very quickly. (That's why you're not supposed to open up a TV set: it has a lot of capacitors that hold high voltages.) How it works, briefly, is that two conductive surfaces are separated by an insulator. If the electrodes

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